

ANKITHA SANNAMALLAPPA		asannamal@ltu.edu
Mechanical Design Engineering		+1-409-294-6159
		Ann Arbor, MI, USA
Experienced Mechanical Design Engineer skilled in CAD modeling, testing, and data analysis for system performance enhancement. Proficient in planning, documentation, and compliance with safety/regulatory standards. Specializes in reliable and cost-effective mechanical systems.		
Work Experience		
Mechanical Engineering		Nov 2024 - Mar 2025
Caresoft Global		
- Conducted reliability analysis of hardware components, meticulously documenting failure modes, risk assessments, and benchmarking multiple vehicles to optimize cost and manufacturability for mass production car lines while adhering to testing schedules. - Performed failure and stress analyses, implemented design improvements based on rigorous test data, and prepared detailed reports for supervisor review to support reliable mechanical designs. - Utilized CAD modeling techniques in design experiments to develop innovative mechanical systems that enhanced efficiency and reduced costs. Implemented manufacturing engineering solutions to optimize production processes and reduce operational costs. - Designed and developed mechanical systems for product engineering projects, ensuring reliable performance under varied test conditions. - Analyzed testing results to drive engineering design improvements, integrating failure analysis data and stress testing insights. - Developed innovative equipment design solutions for industrial applications, leading to improved operational efficiency. - Formulated and executed engineering development plans that enhanced mechanical design and manufacturing processes. - Created mechanical systems presentations for client discussions, ensuring technical clarity and effective communication. - Engineered high-performance motors for industrial applications, contributing to increased system efficiency and productivity. - Developed program management strategies to enhance project delivery and team performance through coordinated scheduling and reporting. - Collaborated with stakeholders to identify and prioritize project requirements, resulting in improved project outcomes. - Managed data collection and analysis for healthcare information systems, leveraging analytical skills to support strategic planning.		
Product Design Engineer		May 2023 - Aug 2023
EZ Metrology		
- Designed using CATIA V5 and developed technical documentation and test setups for hardware components, ensuring compliance with precision and manufacturability standards while following structured test processes. - Conducted prototype reliability testing and performance evaluations, refining test procedures with calibrated jigs to enhance component durability. - Optimized mechanical component designs by implementing data analysis software, achieving a 20% improvement in efficiency through systematic testing. - Streamlined engineering processes by incorporating Python programming, enhancing testing efficiency and output quality. - Integrated cross-functional collaboration in testing and system evaluations to ensure robustness and reliability in design outcomes. - Developed and executed metrology systems for quality control, rigorously testing components to meet precision standards.		
Mechanical Design Engineer		Aug 2022 - May 2023
Ace Manufacturing Systems		
- Designed and developed mechanical systems using AutoCAD, which improved product efficiency and reduced production costs through effective test planning and reporting. - Established reliability test systems and methodologies to verify compliance with performance, durability, and safety requirements and communicated findings with supervisory teams. - Applied firmware knowledge in troubleshooting mechanical equipment, ensuring prompt resolution of test-related issues and increased productivity. Integrated stamping processes into production workflows to optimize quality control measures and support scheduled calibrations. Engineered tooling solutions to improve manufacturing efficiency, supported by precise test data and performance reporting. - Demonstrated proficiency with extrusion processes to enhance operational efficiency and align with systematic testing frameworks. - Implemented material handling systems that reduced labor costs and increased productivity, incorporating test feedback into continuous improvement initiatives. - Developed simulations and models for mechanical systems using coding techniques, enabling data-driven design adjustments and calibration verifications. - Communicated technical designs and testing outcomes effectively to cross-functional teams and stakeholders through strong presentation skills. Utilized analytical skills to design innovative mechanical systems that improved efficiency and reduced costs, with a focus on scheduled test processes. Designed and developed mechanical systems that operated within environmental restraints and packaging to ensure optimal performance and efficiency - Optimized manufacturing operations through systematic improvements, ensuring production alignment with stringent test protocols and reporting standards.		
Core Skills		
Testing & Analysis : Failure Analysis, Root Cause Analysis, Design of Experiments (DOE), Rapid Prototyping & Simulation : ANSYS Workbench, Test Prototype, Structural Integrity Testing, Reliability Testing, Root Cause Failure Analysis, DFMEA, Design Optimization, Simulation & CAD : ANSYS Workbench for Structural and Thermal Analysis, CFD, Structural & Thermal FEA, CAD & Drafting : SolidWorks, CATIA V5, AutoCAD, Advanced GD&T, High-Precision 3D CAD Model, 3D Modeling, manufacturing methods, CAE, excellent organizational skills, Problem Solving, written skills, articulate, motivated, Flexible, regulatory requirements, R&D, PPO		
Education		
Lawrence Technological University		Aug 2023 - May 2025
Master of Engineering - Mechanical Engineering		
Global Academy of Technology		Jun 2018 - Aug 2022
Bachelor of Engineering - Mechanical Engineering		